



Discord Bans r/WallStreetBets

Discord has banned the server used by r/WallStreetBets over "hate speech" and glorifying violence. <http://digit.in/feb21-41>



Birdwatch

Twitter has launched "Birdwatch" a community-based programme to battle misinformation. <http://digit.in/feb21-42>



RADAR

A look into the history and evolution of radar technology

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adar technology has grown by leaps and bounds from the time of its inception. Radar has been an invaluable

military tool for decades. A radar system typically involves transmitting a high-frequency signal towards the location of a target (or even in multiple directions) and receiving reflected signals that pinpoint the position of said target. The returning signal is processed and information can be extracted regarding the target, its position, and speed. The technology

once only served military use cases, but now it is finding its place in commercial, industrial, medical, weather, and automotive systems as well. Let's delve into the history of radar technology and see how it evolved through the years.

EARLY BEGINNINGS

Although serious developmental work on radar technology began in the 1930s, the idea of radar had its origins in the experiments on electromagnetic radiation that were conducted by German

physicist, Heinrich Hertz during the 1880s. Hertz formulated equations of the electromagnetic field that determined that light and radio waves both had the same fundamental laws but had widely differing frequencies. His work led to the deduction that radio waves can be reflected from metallic objects and refracted by a dielectric medium, just like light waves.

Based on the principles outlined by Hertz, Christian Hülsmeyer, a German engineer, carried out experiments using radar waves. While experimenting with Hertz's reflector, Hülsmeyer noticed that electromagnetic waves emitted from a transmitter and reflected by a metallic surface could be utilised to detect distant metallic objects and obstacles. He worked on developing a device that could measure the transit time of the reflected waves. The device was dubbed the telemobiloscope. In 1904, Hülsmeyer filed a patent surrounding this device and was granted the patent in several countries. The device was used in ships as an obstacle detector and navigation aid.

In 1936, Robert Page, who later became the Director of Research for the US Naval Research Laboratory, demonstrated the pulse radar. These systems were able to detect continuous wave patterns generated by ship and aircraft engines. These patterns of pulses were timed on an oscilloscope and then the direction of the antenna would reveal the angular location of targets as well as their range. By identifying these two variables, the target could ultimately be pinpointed. As a result of this work, Page, along with Dr Alfred Taylor and Dr Leo C. Young, were credited with authenticating the world's first radar.



"Telemobiloscope" by Christian Hülsmeyer, 1904

Image source: Deutsches Museum